# Lab Assignment 03

BRAC UNIVERSITY

Inspiring Excellence

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|---|---|
| **Course Code:** | **CSE111** |
| **Course Title:** | **Programming Language II** |
| **Topic:** | **Method Overloading, Constructor & Overloading** |
| **Number of Tasks:** | **12 (Classwork: 06, Homework: 06)** |

***[Submit all the Coding Tasks (Homework: Task 1 to 4) in the Google Form shared on buX before the next lab. Submit the Tracing Tasks (Homework: Task 5 to 6) handwritten to your Lab Instructors at the beginning of the lab]***

**[You are not allowed to change the driver codes of any of the tasks]**

# CLASSWORK

## Task 1

Design the **Shape** class with necessary properties to produce the given output for the provided driver code.

| Driver Class | Output |
|---|---|
| `public class ShapeTester{`<br>`  public static void main(String args []){`<br>`  Shape circle = new Shape();`<br>`  Shape triangle = new Shape();`<br>`  Shape rectangle = new Shape();`<br><br>`  circle.setParameters("Circle", 5);`<br>`  triangle.setParameters("Triangle", 4, 7);`<br>`  rectangle.setParameters("Rectangle", 2.4, 4.4);`<br><br>`  System.out.println(circle.details());`<br>`  System.out.println("1---------------");`<br>`  System.out.println(triangle.details());`<br>`  System.out.println("2---------------");`<br>`  System.out.println(rectangle.details());`<br>`  }`<br>`}` | `Shape Name: Circle`<br>`Area: 78.54`<br>`1---------------`<br>`Shape Name: Triangle`<br>`Area: 14.0`<br>`2---------------`<br>`Shape Name: Rectangle`<br>`Area: 10.56` |

## Task 2

Design the **Student** class in such a way that it produces the following output.

| Driver Code | Expected Output |
|---|---|
| `public class StudentTester{`<br>`  public static void main(String[] args){`<br>`  Student s1 = new Student("Harry", "CSE");`<br>`  System.out.println(s1.name);`<br>`  s1.updateName("Harry Potter");`<br>`  System.out.println(s1.name);`<br>`  System.out.println(s1.prog);`<br>`  s1.updateProgram("CS");`<br>`  String var = s1.accessProgram();`<br>`  System.out.println(var);`<br>`  }`<br>`}` | `Harry`<br>`Harry Potter`<br>`CSE`<br>`CS` |

# Task 3

Design the **Shape2D** class in such a way that it produces the following output.

| Driver Code | Expected Output |
|---|---|
| `public class Shape2DTester {`<br>`  public static void main(String[] args) {`<br>`  Shape2D sq = new Shape2D();`<br>`  System.out.println("---------1----------");`<br>`  sq.area();`<br>`  System.out.println("---------2----------");`<br>`  Shape2D rectangle = new Shape2D(5, 6);`<br>`  System.out.println("---------3----------");`<br>`  rectangle.area();`<br>`  System.out.println("---------4----------");`<br>`  Shape2D tri1 = new Shape2D(5, 6, "Triangle");`<br>`  System.out.println("---------5----------");`<br>`  tri1.area();`<br>`  System.out.println("---------6----------");`<br>`  Shape2D tri2 = new Shape2D(5, 6, 7);`<br>`  System.out.println("---------7----------");`<br>`  tri2.area();`<br>`  }`<br>`}` | `A Square has been created with length: 5`<br>`---------1----------`<br>`The area of the Square is: 25.00`<br>`---------2----------`<br>`A Rectangle has been created with length: 5 and breath:  6`<br>`---------3----------`<br>`The area of the Rectangle is: 30.00`<br>`---------4----------`<br>`A Triangle has been created with height: 5 and base:  6`<br>`---------5----------`<br>`The area of the Triangle is: 15.00`<br>`---------6----------`<br>`A Triangle has been created with the following sides: 5, 6, 7`<br>`---------7----------`<br>`The area of the Triangle is: 14.70` |

# Task 4

Design the **Library** class with the necessary properties so that the given output is produced for the provided driver code.

| Driver Code | Output |
|---|---|
| `public class Tester5{`<br>`  public static void main(String[] args) {`<br>`  Library a1 = new Library(3);`<br>`  System.out.println("1----------");`<br>`  a1.addBook("Ice");`<br>`  System.out.println("2----------");`<br>`  a1.printDetail();`<br>`  System.out.println("3----------");`<br>`  a1.addBook("Emma");`<br>`  a1.addBook("Wings");`<br>`  a1.addBook("Next");`<br>`  System.out.println("4----------");`<br>`  a1.printDetail();`<br>`  System.out.println("5----------");`<br>`  Library a2 = new Library(4);`<br>`  System.out.println("6----------");`<br>`  a2.addBook("Onnobhubon");`<br>`  a2.addBook("Ami");`<br>`  System.out.println("7----------");`<br>`  a2.printDetail();`<br>`  System.out.println("8----------");`<br>`  a2.addBook("Deyal");`<br>`  a2.addBook("Himu");`<br>`  a2.addBook("Megher Upor Bari");`<br>`  System.out.println("9----------");`<br>`  a2.printDetail();`<br>`  }`<br>`}` | A library has been created with capacity 3<br>1----------<br>Book 'Ice' added to the library<br>2----------<br>Maximum Capacity: 3<br>Total Books: 1<br>Book list:<br>Ice<br>3----------<br>Book 'Emma' added to the library<br>Book 'Wings' added to the library<br>Exceeds maximum capacity. You can't add more than 3 books<br>4----------<br>Maximum Capacity: 3<br>Total Books: 3<br>Book list:<br>Ice<br>Emma<br>Wings<br>5----------<br>A library has been created with capacity 4<br>6----------<br>Book 'Onnobhubon' added to the library<br>Book 'Ami' added to the library<br>7----------<br>Maximum Capacity: 4<br>Total Books: 2<br>Book list:<br>Onnobhubon<br>Ami<br>8----------<br>Book 'Deyal' added to the library<br>Book 'Himu' added to the library<br>Exceeds maximum capacity. You can't add more than 4 books<br>9----------<br>Maximum Capacity: 4<br>Total Books: 4<br>Book list:<br>Onnobhubon<br>Ami<br>Deyal<br>Himu |

# Task 5

Design the **TaxiLagbe** class with necessary properties to produce the given output for the provided driver code.

**Driver Code**

```
public class TaxiTester{
 public static void main(String[] args) {
   TaxiLagbe taxi1 = new TaxiLagbe("1010-01","Dhaka");
   System.out.println("1-------------");
   taxi1.printDetails();
   System.out.println("2-------------");
   taxi1.addPassenger("Wilson", 105);
   System.out.println("3-------------");
   taxi1.printDetails();
   System.out.println("4-------------");
   taxi1.addPassenger("Walker", 100, "Wood", 200);
   System.out.println("5-------------");
   taxi1.printDetails();
   System.out.println("6-------------");
   taxi1.addPassenger("Karen", 200);
   taxi1.addPassenger("Donald", 130);
   System.out.println("7-------------");
   taxi1.printDetails();
   System.out.println("8-------------");
   TaxiLagbe taxi2 = new TaxiLagbe("1010-02");
   System.out.println("9-------------");
   taxi2.addPassenger("Don", 115, "Parker", 215);
   System.out.println("10-------------");
   taxi2.printDetails();
 }
}
```

**Output**

```
A new taxi has been created
1-------------
Taxi number: 1010-01
This taxi can cover Dhaka area
Total Passenger: 0
Passenger List:
Total collected fare: 0 Taka
2-------------
Dear Wilson! Welcome to TaxiLagbe
3-------------
Taxi number: 1010-01
This taxi can cover Dhaka area
Total Passenger: 1
Passenger List:
Wilson
Total collected fare: 105 Taka
4-------------
Dear Walker! Welcome to TaxiLagbe
Dear Wood! Welcome to TaxiLagbe
5-------------
Taxi number: 1010-01
This taxi can cover Dhaka area
Total Passenger: 3
Passenger List:
Wilson Walker Wood
Total collected fare: 405 Taka
6-------------
Dear Karen! Welcome to TaxiLagbe
Taxi Full! No more passengers can be
added
7-------------
Taxi number: 1010-01
This taxi can cover Dhaka area
Total Passenger: 4
Passenger List:
Wilson Walker Wood Karen
Total collected fare: 605 Taka
8-------------
A new taxi has been created
9-------------
Dear Don! Welcome to TaxiLagbe
Dear Parker! Welcome to TaxiLagbe
10-------------
Taxi number: 1010-02
This taxi can cover Khulna area
Total Passenger: 2
Passenger List:
Don Parker
Total collected fare: 330 Taka
```

# Task 6

```
1  public class A{
2    public int temp = 3, sum = 9, y = 4, x = 0;
3    public A(){
4      int sum = 7;
5      y = temp - 5;
6      sum = temp + 2;
7      temp -= 2;
8      this.x = sum + temp + y;
9    }
10   public A(int y, int temp){
11     y = temp - 1 + x;
12     sum = temp + 2 -x;
13     temp-=2;
14   }
15   public void methodB(int m, int [] n){
16     int x = 0;
17     y = y + m + this.methodB(x,m);
18     x = this.x + 2 + (++n[0]);
19     sum = sum + x + y;
20     n[0] = sum + 2;
21     System.out.println(n[0] + " " + y+ " " + sum);
22   }
23   public int methodB(int m, int n){
24     int [] y = {10};
25     this.y = y[0] + this.y + m;
26     x = this.y + 2 + temp - n;
27     sum = x + y[0] + this.sum;
28     System.out.println(y[0]+ " "+ temp + " " + sum);
29     return y[0];
30   }
31 }
```

| Driver Code | Output | | |
|---|---|---|---|
| public class Tester6 {<br>public static void main(String args[]){<br>int[] x = {35};<br>A a1;<br>a1 = new A();<br>A a2 = new A(-5,-7);<br>a1.methodB(1, x);<br>a2.methodB(1, x);<br>}<br>} | | | |
| | | | |
| | | | |
| | | | |

# HOMEWORK

## Task 1

Design the **Toy** class in such a way that it produces the following output

| Driver Code | Expected Output |
|---|---|
| `public class ToyTester{`<br>`  public static void main(String[] args){`<br>`  Toy t1 = new Toy("Car", 230);`<br>`  System.out.println("1==========");`<br>`  t1.updatePrice(250);`<br>`  System.out.println("2==========");`<br>`  System.out.println(t1.name);`<br>`  t1.showPrice();`<br>`  System.out.println("3==========");`<br>`  Toy t2 = new Toy("Robot", 450);`<br>`  System.out.println("4==========");`<br>`  t2.updateName("Autobot");`<br>`  System.out.println("5==========");`<br>`  System.out.println(t2.name);`<br>`  t2.showPrice();`<br>`  }`<br>`}` | `A new toy has been made!`<br>`1==========`<br>`2==========`<br>`Car`<br>`price: 250 Taka`<br>`3==========`<br>`A new toy has been made!`<br>`4==========`<br>`Changing old name: Robot`<br>`new name: Autobot`<br>`5==========`<br>`Autobot`<br>`price: 450 Taka` |

## Task 2

Complete the following **Cart** class to generate the given output from the tester code:

- A cart will have a cart number which will be assigned in ***create_cart()*** method.
- Each cart can hold up to 3 items (at max).
- Each cart must have two arrays to store items and their respective prices.
- The items inside a cart will be added in ***addItem()*** method only if the cart items do not exceed 3.
- The ***giveDiscount()*** method saves the discount given to that cart object and updates the price accordingly.

| Driver Code | Output |
|---|---|
| `public class CartTester{`<br>`  public static void main(String [] args){`<br>`  Cart c1 = new Cart ();`<br>`  Cart c2 = new Cart ();`<br>`  Cart c3 = new Cart ();`<br><br>`  c1.create_cart(1);`<br>`  c2.create_cart(2);`<br>`  c3.create_cart(3);`<br>`  System.out.println("====1====");`<br>`  c1.addItem("Table", 3900.5);`<br>`  c1.addItem("Chair", 1400.76);`<br>`  c1.addItem(5400.87, "Television");`<br>`  c1.addItem(5000.0, "Refrigerator");`<br><br>`  System.out.println("====2====");`<br>`  c2.addItem("Stove",439.90);`<br><br>`  System.out.println("====3====");`<br>`  c3.addItem("Chair",1400.5);`<br>`  c3.addItem(3400.0, "Chair");`<br><br>`  System.out.println("====4====");`<br>`  c1.cartDetails();`<br><br>`  System.out.println("====5====");`<br>`  c2.cartDetails();`<br><br>`  System.out.println("====6====");`<br>`  c3.cartDetails();`<br>`  c1.giveDiscount(10);`<br><br>`  System.out.println("====7====");`<br>`  c1.cartDetails();`<br>`  }`<br>`}` | `====1====`<br>`Table added to cart 1.`<br>`You have 1 item(s) in your cart now.`<br>`Chair added to cart 1.`<br>`You have 2 item(s) in your cart now.`<br>`Television added to cart 1.`<br>`You have 3 item(s) in your cart now.`<br>`You already have 3 items on your cart`<br>`====2====`<br>`Stove added to cart 2.`<br>`You have 1 item(s) in your cart now.`<br>`====3====`<br>`Chair added to cart 3.`<br>`You have 1 item(s) in your cart now.`<br>`Chair added to cart 3.`<br>`You have 2 item(s) in your cart now.`<br>`====4====`<br>`Your cart(c1) :`<br>`Table - 3900.5`<br>`Chair - 1400.76`<br>`Television - 5400.87`<br>`Discount Applied: 0.0%`<br>`Total price: 10702.130000000001`<br>`====5====`<br>`Your cart(c2) :`<br>`Stove - 439.9`<br>`Discount Applied: 0.0%`<br>`Total price: 439.9`<br>`====6====`<br>`Your cart(c3) :`<br>`Chair - 1400.5`<br>`Chair - 3400.0`<br>`Discount Applied: 0.0%`<br>`Total price: 4800.5`<br>`====7====`<br>`Your cart(c1) :`<br>`Table - 3900.5`<br>`Chair - 1400.76`<br>`Television - 5400.87`<br>`Discount Applied: 10.0%`<br>`Total price: 9631.917000000001` |

# Task 3

Design the **Reader** class in such a way so that the following code provides the expected output.

- A reader will have a name, capacity to read and an array of books they are reading.
- The initial capacity of a reader will be 2. The initial name will be "New user".

**Driver Code**

```
public class Reader_tester {
  public static void main(String[] args){

    System.out.println("1 ==========");
    Reader r1 = new Reader("Messi");
    Reader r2 = new Reader("Ronaldo", 3);
    System.out.println("2 ==========");
    r1.readerInfo();
    System.out.println("3 ==========");
    r2.addBook("Java");
    r2.addBook("Python");
    r2.addBook("C++");
    r2.readerInfo();
    System.out.println("4 ==========");
    r1.addBook("C#");
    r1.addBook("Rust");
    r1.addBook("GoLang");
    System.out.println("5 ==========");
    r2.addBook("Python");
    System.out.println("6 ==========");
    r1.readerInfo();
    System.out.println("7 ==========");
    r1.updateCapacity(3);
    System.out.println("8 ==========");
    r1.addBook("GoLang");
    System.out.println("9 ==========");
    r1.readerInfo();

  }
}
```

**Expected Output**

```
1 ==========
A new reader is created!
A new reader is created!
2 ==========
Name: Messi
Capacity: 2
Books:
No books added yet
3 ==========
Name: Ronaldo
Capacity: 3
Books:
Book 1: Java
Book 2: Python
Book 3: C++
4 ==========
No more capacity
5 ==========
No more capacity
6 ==========
Name: Messi
Capacity: 2
Books:
Book 1: C#
Book 2: Rust
7 ==========
Capacity has changed to 3
8 ==========
9 ==========
Name: Messi
Capacity: 3
Books:
Book 1: C#
Book 2: Rust
Book 3: GoLang
```

# Task 4

You are building a tracker system that will keep track of a person's income and expenses.

- When the ***createTracker()*** method is invoked it sets the balance to 1.0 taka.
- The ***info()*** method **returns** a String with the trackers information.
- If the total balance becomes 0 after the ***expense()*** method is called it prints "You're broke!" . Again if the available balance is less than the expense it prints "Not enough balance.". Otherwise the method prints "Balance updated" after updating the balance.
- The last expense and income history can be seen by using the ***history()*** method.

**Driver Code**

```
public class Tester4{
 public static void main(String[] args) {
    MoneyTracker tr1 = new MoneyTracker();
    System.out.println(tr1.info());
    tr1.createTracker("John");
    System.out.println("1 ==========");
    System.out.println(tr1.info());
    System.out.println("2 ==========");
    tr1.income(1000);
    System.out.println(tr1.info());
    System.out.println("3 ==========");
    tr1.expense(800);
    tr1.expense(100);
    System.out.println(tr1.info());
    System.out.println("4 ==========");
    tr1.showHistory();
    System.out.println("5 ==========");
    tr1.expense(101);
    System.out.println("6 ==========");
    tr1.expense(200);
    System.out.println("7 ==========");
    tr1.income(200);
    tr1.showHistory();
    System.out.println("8 ==========");
 }
}
```

**Output**

```
A new money tracker has been launched.
Name: null
Current Balance: 0.0
1 ==========
Name: John
Current Balance: 1.0
2 ==========
Balance Updated!
Name: John
Current Balance: 1001.0
3 ==========
Balance Updated.
Balance Updated.
Name: John
Current Balance: 101.0
4 ==========
Last added: 1000.0
Last spent: 100.0
5 ==========
You're broke!
6 ==========
Not enough balance.
7 ==========
Balance Updated!
Last added: 200.0
Last spent: 101.0
8 ==========
```

# Task 5

```
public class A{
    public int x, y, z = 5;
    public double p = 0.0;
    public void methodA(int x, int m) {
        this.x = methodB(this.x);
        p = x + this.x % m * 3.0;
        y = y + methodB(x++, this.x);
        System.out.println(this.x +" " + x + y + " " + p) ;
    }
    public int methodB(int q, int n) {
        int arr[] = {3,4,5};
        arr[0] = arr[0] + this.x + n;
        arr[1] = q + arr[1];
        System.out.println(arr[0] +" " + arr[1] + " " + arr[2]) ;
        return arr[1] + arr[2];
    }
    public int methodB(int y) {
        if(y % 2 == 0) {
            int temp = this.methodB(2, y);
            return temp;
        }
        else{
            return 4;
       }
    }
}
```

| Driver Code | Output | | |
|---|---|---|---|
| `public class Tester11 {`<br>`  public static void main(String [] args){`<br>`  A a1 = new A();`<br>`  a1.methodA(2,3);`<br>`  a1.methodB(5,4);`<br>`  }`<br>`}` | | | |
| | | | |
| | | | |
| | | | |

# Task 6

```
1  public class Maze{
2          public int x;
3          public Maze(){
4            int x = 17;
5            this.methodB(x, -this.x);
6          }
7          public void methodA(){
8              int m = 0, x = 9;
9              m = methodB(m-3)+x;
10             this.x = ++x;
11             System.out.println(this.x+" "+m);
12             this.methodB(x,m);
13             System.out.println(x+" "+(m+this.x));
14             methodB(m);
15         }
16         public int methodB(int y){
17             x=y*y;
18             System.out.println(x+" "+y);
19             return x-11;
20         }
21         public void methodB(int z, int x){
22             z=z-2;
23             x=this.x-2*x;
24             System.out.println(z+" "+this.x);
25         }
26 }
```

| DRIVER CODE | OUTPUTS | |
|---|---|---|
| public class MazeTester{ | | |
| public static void main(String args []){ | | |
| Maze m1 = new Maze(); | | |
| m1.methodA(); | | |
| new Maze().methodB(7,15); | | |
| } | | |
| } | | |
| | | |
| | | |

# Ungraded Tasks (Optional)

(You don't have to submit the ungraded tasks)

## Task 1

| Driver Code | Output |
|---|---|
| `public class CustomerTester {`<br>`  public static void main(String[] args) {`<br>`  Customer c1 = new Customer();`<br>`  c1.createCustomer("John");`<br>`  System.out.println("1====================");`<br>`  c1.showCart();`<br>`  System.out.println("2====================");`<br>`  c1.addItem("Apple", 2);`<br>`  c1.addItem("Bread", 5);`<br>`  c1.addItem("Milk");`<br>`  c1.addItem("Eggs", 2);`<br>`  System.out.println("3====================");`<br>`  c1.showCart();`<br>`  System.out.println("4====================");`<br>`  c1.calculatePrice();`<br>`  System.out.println("5====================");`<br>`  Customer c2 = new Customer();`<br>`  c2.createCustomer("Jane");`<br>`  c2.addItem("Apple", 2, "Orange", 5);`<br>`  c2.addItem("Chocolates", 15, "Bread", 5);`<br>`  c2.addItem("Milk", 3);`<br>`  System.out.println("6====================");`<br>`  c2.showCart();`<br>`  System.out.println("7====================");`<br>`  c2.calculatePrice();`<br>`  }`<br>`}` | `1====================`<br>`Customer: John`<br>`2====================`<br>`Apple added to cart`<br>`Orange added to cart`<br>`Bread added to cart`<br>`Milk added to cart`<br>`Cart is full`<br>`3====================`<br>`Customer: John`<br>`Item: Apple Price: 2`<br>`Item: Orange Price: 5`<br>`Item: Bread Price: 5`<br>`Item: Milk Price: 3`<br>`4====================`<br>`Total: 15`<br>`5====================`<br>`Apple and Orange added to cart`<br>`Chocolates and Bread added to cart`<br>`Cart is full`<br>`6====================`<br>`Customer: Jane`<br>`Item: Apple Price: 2`<br>`Item: Orange Price: 5`<br>`Item: Chocolates Price: 15`<br>`Item: Bread Price: 5`<br>`7====================`<br>`Total: 27` |

## Task 2

| Driver Code | Sample Output |
|---|---|
| `public class CalculatorTester {`<br>`  public static void main(String[] args) {`<br>`  Calculator calc = new Calculator();`<br><br>`  System.out.println("1==========");`<br>`  calc.add(10, 20);`<br>`  System.out.println("2==========");`<br>`  calc.add(5, 15, 25);`<br>`  System.out.println("3==========");`<br>`  calc.multiply(6, 7);`<br>`  System.out.println("4==========");`<br>`  calc.multiply(2, 3, 4);`<br>`  System.out.println("5==========");`<br>`  calc.multiply("Hello", 3);`<br>`  System.out.println("6==========");`<br>`  calc.multiply("Java", 5);`<br>`  }`<br>`}` | `1==========`<br>`30`<br>`2==========`<br>`45`<br>`3==========`<br>`42`<br>`4==========`<br>`24`<br>`5==========`<br>`Hello-Hello-Hello`<br>`6==========`<br>`Java-Java-Java-Java-Java` |

## Task 3

| Driver Code | Sample Output |
|---|---|
| `public class LibraryTest {`<br>`  public static void main(String[] args) {`<br><br>`  Book book1 = new Book();`<br>`  book1.createBook("The Great Gatsby");`<br><br>`  Book book2 = new Book();`<br>`  book2.createBook("1984", "George Orwell");`<br><br>`  Book book3 = new Book();`<br>`  book3.createBook("To Kill a Mockingbird", "Harper Lee", "Fiction");`<br><br>`  System.out.println("\|---Book Customization---\|");`<br>`  book1.customizeGenre("Classic");`<br>`  book1.customizePages(180);`<br><br>`  book2.customizeGenre("Dystopian");`<br>`  book2.customizePages(328);`<br><br>`  book3.customizePages(281);`<br><br>`  System.out.println();`<br><br>`  System.out.println("\|---Library Inventory---\|");`<br>`  book1.displayDetails();`<br>`  book2.displayDetails();`<br>`  book3.displayDetails();`<br>`  }`<br>`}` | `\|---Book Customization---\|`<br>`Updated genre of "The Great Gatsby" to Classic.`<br>`Updated pages of "The Great Gatsby" to 180 pages.`<br>`Updated genre of "1984" to Dystopian.`<br>`Updated pages of "1984" to 328 pages.`<br>`Updated pages of "To Kill a Mockingbird" to 281 pages.`<br><br>`\|---Library Inventory---\|`<br>`Title: The Great Gatsby, Author: Unknown, Genre: Classic, Pages: 180`<br>`Title: 1984, Author: George Orwell, Genre: Dystopian, Pages: 328`<br>`Title: To Kill a Mockingbird, Author: Harper Lee, Genre: Fiction, Pages: 281` |

# Task 4

| Driver Code | Sample Output |
|---|---|
| `public class MovieManagerTest {`<br>`  public static void main(String[] args) {`<br>`  Movie inception = new Movie();`<br>`  inception.setMovieDetails("Inception", "Christopher Nolan", 8.8);`<br><br>`  System.out.println("1====================");`<br>`  inception.addActors("Leonardo DiCaprio", "Joseph Gordon-Levitt");`<br>`  inception.addActors("Ellen Page");`<br>`  inception.showInfo();`<br><br>`  System.out.println("2====================");`<br>`  Movie avengers = new Movie();`<br>`  avengers.setMovieDetails("Avengers: Endgame", "Anthony Russo", 8.4);`<br>`  avengers.addActors("Robert Downey Jr.", "Chris Evans", "Scarlett Johansson");`<br>`  avengers.showInfo();`<br><br>`  System.out.println("3====================");`<br>`  Movie parasite = new Movie();`<br>`  parasite.setMovieDetails("Parasite", "Bong Joon-ho");`<br>`  parasite.addActors("Song Kang-ho", "Choi Woo-shik");`<br>`  parasite.updateRating(8.6);`<br>`  parasite.showInfo();`<br><br>`  System.out.println("4====================");`<br>`  parasite.updateRating(8.9);`<br>`  parasite.showInfo();`<br>`  }`<br>`}` | `1====================`<br>`Added actor "Leonardo DiCaprio" to "Inception".`<br>`Added actor "Joseph Gordon-Levitt" to "Inception".`<br>`Added actor "Ellen Page" to "Inception".`<br>`Title: Inception`<br>`Director: Christopher Nolan`<br>`Rating: 8.8`<br>`Actors: Leonardo DiCaprio, Joseph Gordon-Levitt, Ellen Page`<br>`2====================`<br>`Added actor "Robert Downey Jr." to "Avengers: Endgame".`<br>`Added actor "Chris Evans" to "Avengers: Endgame".`<br>`Added actor "Scarlett Johansson" to "Avengers: Endgame".`<br>`Title: Avengers: Endgame`<br>`Director: Anthony Russo`<br>`Rating: 8.4`<br>`Actors: Robert Downey Jr., Chris Evans, Scarlett Johansson`<br>`3====================`<br>`Added actor "Song Kang-ho" to "Parasite".`<br>`Added actor "Choi Woo-shik" to "Parasite".`<br>`Updated rating of "Parasite" to 8.6`<br>`Title: Parasite`<br>`Director: Bong Joon-ho`<br>`Rating: 8.6`<br>`Actors: Song Kang-ho, Choi Woo-shik`<br>`4====================`<br>`Updated rating of "Parasite" to 8.9`<br>`Title: Parasite`<br>`Director: Bong Joon-ho`<br>`Rating: 8.9`<br>`Actors: Song Kang-ho, Choi Woo-shik` |

# Task 5

```
public class Test3 {
    int x = 2, y = 4, sum = 3;
    int arr[] = {x, y, sum};
    public void methodA(int x) {
        arr[0] += methodB(y, this.x) + methodC(x);
        System.out.println(x + " " + this.x + " " + sum);
        arr[1] += this.x * (++y) / (sum % x);
        System.out.println(y + " " + sum + " " + this.x);
        arr[2] += methodC(x) + methodB(this.x, sum);
        System.out.println(arr[0] + " " + arr[1] + " " + arr[2]);
    }
    public int methodB(int q, int n) {
        int arr2[] = {7, 8};
        int a = (arr2[0]++) - q;
        int b = (++arr2[1]) - n;
        return a + b;
    }
    public int methodC(int z) {
        z = sum + methodB(x, sum) - z;
        return z/2;
    }
}
```

```
public class Tester3{
    public static void main(String [] args){
        Test3 t3 = new Test3();
        t3.methodA(7);
    }
}
```

| Outputs | | |
|---|---|---|
| | | |
| | | |
| | | |